

Homework _num₂

Claire Trebing, Jason Young, Illya Starikov

February 25, 2016

Question _num₁

- a. Valid Operation
- b. Valid Operation
- c. Key Constraint (*Key_num₄isalreadytaken*)*KeyConstraint, NotNullConstraint(Keycannotbenull)*
- d. Valid Operation
- e. Valid Operation
- f. Integrity Constraint (Did not update WORKS_ON or DEPENDANT)
- g. Integrity Constraint (Did not update WORK_ON)
- h. Valid Operation
- i. Integrity Constraint (Did not update WORKS_ON)
- j. Valid Operation

Question _num₂

Grader note: the rho (ρ) is too small in L^AT_EX to be distinguished from a subscript, so instead P is used in it's place.

- a. $\Pi_{\text{Fname}, \text{Minit}, \text{Lname}}(\sigma_{\text{Dno} = '5' \wedge \text{hours} = >10}(\text{Project} \bowtie_{\text{Dname} = 'ProjectX'} \text{Works_On}))$
- b. $\Pi_{\text{Fname} = \text{Dependent_Name}}(\text{Employee} \bowtie \text{Dependent})$
- c. $\Pi_{\text{Fname}, \text{LName}}(\text{Employee} \div \Pi_{\text{Super_SSN}}(\sigma_{\text{Fname} = 'Frankline' \wedge \text{Lname} = 'Wong'} \text{DEPARTMENT}))$
- d. $\Pi_{\text{Lname}, \text{Sum_Hours}}(s_{\text{name}} \mathcal{F}_{\text{sum hours}}(\text{PROJECT} \bowtie_{\rho_{\text{ESSN}} = \text{SSN}} \text{Works_On}))$
- e. $\text{Employee1} \leftarrow \text{Employee} \bowtie_{\rho_{\text{ESSN}} = \text{SSN}} \text{Work_On}$
 $\Pi_{\text{Lname}, \text{PNo}} \text{Employee1} \div \Pi_{\text{Pno}} \text{Employee1}$
- f. $\Pi_{\text{Lname}}(\Pi_{\text{Lname}, \text{SSN}} \text{Employee} - \Pi_{\text{Lname}, \text{SSN}}(\text{Employee} \bowtie_{\rho_{\text{ESSN}} = \text{SSN}} \text{Works_On}))$

- g. $\Pi_{\text{Fname, Average Salary}}(\text{DName} \bowtie \mathcal{F}_{\text{average salary}} \text{P}_{\text{Dno} = \text{Dnumber}} \text{Employee} \bowtie (\text{Department} \bowtie \text{Works_On}))$
- h. $\text{Sex} \mathcal{F}_{\text{average salary}}(\sigma_{\text{sex} = \text{'female'}} \text{Employee})$
- i. $\text{Department1} \leftarrow \text{P}_{\text{DNumber} = \text{DNo}} \text{Project} \bowtie \text{P}_{\text{ESSN} = \text{SSN}} \text{Works_On}$
 $\Pi_{\text{Fname}, \text{Address}}((\text{Employee} \bowtie \text{Department1} - \sigma_{\text{Dlocation} = \text{'Houston'}} \text{Department1}) \bowtie (\sigma_{\text{Dlocation} = \text{'Houston'}} \text{P}_{\text{DNumber} = \text{DNo}}(\text{Department} \bowtie \text{Dept_Locations})))$
- j. $\Pi_{\text{Lname}}(\text{Employee} \bowtie \text{SSN} = \text{MGR_SSN} \text{Department} \bowtie \text{P}_{\text{ESSN} = \text{SSN}}) - \Pi_{\text{Lname}} \text{Employee}$

Question num₃

- Cust# (ORDER)
- Order (ORDER_ITEM, SHIPMENT)
- Item# (ORDER, ITEM)
- Warehouse# (SHIPMENT)

Assumptions

- Table headers with the same name are not unique.